

## Contact

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www.linkedin.com/in/  
arunkumarramanan (LinkedIn)  
deepbrainz.com (Company)  
arunkumarramanan.github.io  
(Portfolio)  
arunkumarblog.medium.com (Blog)

## Top Skills

Machine Learning  
Deep Learning  
Python

## Languages

English (Full Professional)  
Tamil (Native or Bilingual)  
Kannada (Native or Bilingual)  
Hindi (Elementary)  
Telugu (Elementary)

## Certifications

Game Theory, University of British  
Columbia, Stanford University  
Operating Systems  
Introduction to Programming  
(Python)  
Mathematics (Linear Algebra,  
Calculus, Statistics, and Probability)  
Lean Technology Strategy:  
Economic Frameworks for Portfolio  
and Product Management

## Honors-Awards

Yuvasri Kala Bharathi Award  
Best Student of the Year 2006-07  
Kaggle Master  
Won Medal for the rank 62nd/465  
in Google AI's Inclusive Images  
Challenge on Kaggle  
Won Medal for the rank 90th/521 in  
iMet Collection 2019 - FGVC6 on  
Kaggle

# Arunkumar Venkataramanan

Founder, CEO, CPTO & Research Scientist @ DeepBrainz AI &  
Labs | Reasoning & Alignment, Agentic AI & Long-Horizon Autonomy  
Greater Bengaluru Area

## Summary

Research Scientist, Systems Architect, Serial Entrepreneur, and Founder focused on reasoning, alignment, AI safety and reliability, agentic AI, and long-horizon autonomous systems. My work spans foundation models, reasoning systems, agent harnesses, AI infrastructure, and governed autonomous systems designed for reliable real-world deployment. Working across ML engineering, optimization, and large-scale AI systems grounded in mathematical, statistical, and computational reasoning, I build intelligent systems spanning foundation models, reasoning systems, governed agent systems, and agent harnesses that integrate runtime, memory, planning, tool use, verification, and coordinated execution to perform reliable long-horizon work under real-world constraints. As Founder, CEO, and CPTO of DeepBrainz AI & Labs, I lead the development of DeepBrainz-R, Lexopedia AI, and AgentFoundry across model research, training, evals, inference, agent orchestration, autonomous systems, and AI infrastructure. I operate across the full intelligence stack—from foundation model research, training, and evaluation to systems architecture, infrastructure, deployment, and operations. My research focuses on reasoning, memory, planning, alignment, controllability, world models, multi-agent coordination, and long-horizon autonomous systems as foundations for more capable, reliable, and general intelligence. I believe the next frontier will be defined not by model scale alone, but by systems capable of reasoning, planning, acting, verifying, adapting, and operating reliably under real-world constraints and human oversight. Over the past decade, I have worked across AI research, machine learning engineering, distributed systems, infrastructure, product development, and entrepreneurship—translating advances in intelligence research into deployable systems and practical outcomes. My interests span reasoning systems, world models, cognitive architectures, decision-making, software engineering intelligence, human-AI collaboration, reliability, and the foundations of intelligence. Selected affiliations

## Publications

The Product Thinking Playbook

Large Scale Distributed Machine Learning (Deep Learning) Systems

Universal Bot: Conversational AI Chatbot with Generative Language Models and Deep Reinforcement Learning

Universal Language: Natural Language Processing with Python through Deep Learning

Hands-on Generative AI (LLMs) and Deep RL with Python

and communities include NVIDIA Inception, OpenAI Forum, TPU Research Cloud, Microsoft for Startups (Level 5), Google Cloud Technology Partner, AWS Partner, MIT Technology Review Global Panel, Harvard Business Review Ascend, the Python Software Foundation, Kaggle Legacy Master, YC SUS, and F6S Alpha. My mission is to advance reliable, aligned, and governable autonomous systems that expand human capability through intelligent autonomy, practical utility, and long-term value.

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## Experience

### DeepBrainz AI

Founder, CEO, Chief Product and Technologist (CPTO)

November 2019 - Present (6 years 8 months)

Greater Bengaluru Area

- Founder & Chief Architect at DeepBrainz AI, building agent-first AI systems for knowledge work, engineering, and long-horizon autonomous execution.
- Leading DeepBrainz-R, Lexopedia AI, and AgentFoundry across AI research, model development, agent orchestration, infrastructure, deployment, and product strategy.
- Established the core thesis that reliable AI emerges from governed agent systems where reasoning, memory, planning, execution, verification, coordination, and human oversight operate together under real-world constraints, and that reasoning itself is a trainable, controllable capability.
- Architect agent systems that combine models, tools, memory, workflows, governance, and approval mechanisms into production-ready systems capable of sustained work beyond isolated prompts.
- Design efficient AI models and agent runtimes optimized for tool use, structured outputs, long-context reasoning, inference-time scaling, predictable latency, operational economics, and reliability.
- Build governed AI systems where agents perform increasing portions of knowledge and engineering work while humans retain control over objectives, constraints, approvals, risk boundaries, and outcomes.
- Operate across research, systems architecture, engineering, MLOps, AgentOps, infrastructure, and product to translate advanced AI capabilities into deployable systems.
- Shipped and iterated agentic AI systems using deployment feedback, evaluation-driven development, failure analysis, and customer constraints to improve reliability and real-world performance.

- Operate cloud-native AI and ML infrastructure across GCP, AWS, and Azure using vLLM, SGLang, TensorRT, NVIDIA Dynamo, MCP, and A2A.
- Lead platform strategy, inference optimization, commercialization, deployment infra, and go-to-market execution across the DeepBrainz ecosystem.
- Translate frontier AI research into deployable products and defensible infrastructure with measurable reliability, governance, and real-world utility.

## DeepBrainz Labs

### Founding Research Scientist

July 2023 - Present (3 years)

- Conducting frontier AGI and agentic AI research focused on reasoning, memory, planning, coordination, safety, alignment, controllability, long-horizon execution, and superalignment-oriented investigations.
- Designing and training foundation models across LLMs, SLMs, multimodal systems, Transformers, Mixture-of-Experts (MoE), Diffusion models, diffusion-LLMs (DDPM, LDM, DiT, Flow Matching; MDLM, BD3), and hybrid architectures.
- Working across the full training stack including self-supervised pre-training, post-training, instruction tuning, alignment, preference optimization, reinforcement learning (RLHF, RLAIF, RLVR, GRPO), evaluation, safety, inference, inference-time scaling, and test-time scaling.
- Investigating advanced AI architectures including SSM, Mamba, attention-state unification, representation learning systems, world models, controllable reasoning systems, agentic systems, and next-generation foundation model designs.
- Investigating reasoning as a trainable, controllable system capability that can be engineered, evaluated, aligned, and optimized under real-world reliability, safety, latency, and operational constraints.
- Researching representation learning, self-supervised learning, unsupervised learning, reinforcement learning, human-centered AI, memory systems, PEFT, LoRA, efficient adaptation, agent orchestration, and scalable deployment methodologies.
- Building and evaluating AI systems for tool use, multi-agent coordination, software engineering intelligence, autonomous execution, long-horizon task completion, reliability, and real-world performance.
- Utilizing Python, NumPy, PyTorch, JAX, TensorFlow, Keras, scikit-learn, XGBoost, Hugging Face (Transformers, Diffusers, TRL), Megatron-LM, DeepSpeed, NVIDIA NeMo, MLflow, Kubeflow, Ray, and Weights & Biases across large-scale AI research, representation learning, model training,

evaluation, deployment, reasoning systems, agent orchestration, scalable inference, and autonomous execution.

## Stealth Startup

Co-Founder (Research | Engineering | Product)

April 2022 - June 2026 (4 years 3 months)

Greater Bengaluru Area

- Co-founded a consumer AI product in stealth, integrating foundation models, reasoning models, and agentic systems into user-facing intelligent products.
- Led research, engineering, and product design for AI-native workflows spanning knowledge agents, engineering agents, agentic copilots, intelligent search, reasoning, and autonomous execution systems.
- Explored scalable applications of foundation models, reasoning models, agentic systems, and AI-native products across consumer and prosumer domains with a strong focus on usability, adoption, and product-market fit.
- Conducted customer discovery, user interviews, market research, experimentation, validation, and workflow analysis for AI-native products, autonomous engineering systems, and agent-operated software.
- Evaluated opportunities at the intersection of AI agents, software engineering, autonomous systems, coding agents, research workflows, knowledge work, developer tools, and next-generation human–AI collaboration.
- Built and tested hypotheses around AI-native operating systems, agent-operated software, autonomous engineering workflows, intelligent execution systems, knowledge work platforms, and future developer experiences.
- Investigated how specialized agents, memory systems, planning, coordination, tool use, verification, human governance, and software engineering intelligence can improve real-world task completion, multi-agent coordination, and long-horizon efficient execution, informing the foundations of future agentic & autonomous systems.
- Translated advances in foundation models, agent architectures, and autonomous systems into practical products designed to solve meaningful user problems and compound value over time.
- Research, product, and engineering work from this stealth initiative evolved into the foundations of DeepBrainz AI & Labs, including Lexopedia (agentic intelligence for knowledge work), AgentFoundry (governed engineering agents), and the DeepBrainz-R research and model series.

## MIT Technology Review

MIT Technology Review Global Panelist (A panel member)

December 2020 - August 2023 (2 years 9 months)

I volunteered in this role as a panel member

A member of the MIT Technology Review Global Panel, an exclusive forum of thought leaders, innovators, and executives, and have my say on the major tech trends of the day.

As a panelist, I'm asked to take part in executive surveys conducted by MIT Technology Review Insights and share my expertise on a wide range of technology topics and research initiatives.

I receive access to the survey results and specially prepared news and analysis to inform business strategy.

And I have the chance to attend senior-level gatherings held the world over.

<https://www.linkedin.com/groups/13872538>

## Python Software Foundation

A PSF Board Nominee - Contributing Member

January 2020 - February 2022 (2 years 2 months)

A Contributing Member of Python Software Foundation PSF (Official Python) Community. 2X PSF Board Member Nominee.

## Harvard Business Review

HBR Ascend Select Member

February 2020 - January 2021 (1 year)

I was volunteering in this role.

HBR Ascend Select is an exclusive and an invite-only community of aspiring leaders, handpicked by HBR Ascend.

As a member, I get the opportunity to share my views and opinions with the editorial, marketing, and product teams of HBR Ascend.

I also participate in various research projects led by HBR Ascend.

## Stealth Startup

Founder and Research Engineer

June 2016 - October 2019 (3 years 5 months)

Greater Bengaluru Area

- Led early-stage AGI-oriented R&D across deep learning, reinforcement learning, computer vision, NLP, and generative models.
- Transitioned research prototypes into production-grade AI systems, laying the technical foundation for what later became DeepBrainz AI.
- Designed an early AI cloud platform enabling data ingestion, model training, deployment, and operationalization with minimal ML expertise.
- Explored early AGI-oriented research directions that later informed the founding thesis and technical foundations of DeepBrainz AI.

## Independent Software Development Contracts

### Independent Consultant — ML Systems

July 2015 - September 2019 (4 years 3 months)

Greater Bengaluru Area

- Delivered end-to-end AI/ML systems for clients across data engineering, model development, deployment, and product integration.
- Architected distributed machine learning systems and cloud-native platforms for scalable AI applications.
- Advised founders and product teams on ML-driven product strategy, system design, and technical execution.
- Delivered production ML systems spanning computer vision, NLP, recommendation systems, predictive analytics, and cloud-native deployment workflows.

## Kaggle

### Independent ML Engineer (Machine Learning Practitioner)

March 2016 - April 2019 (3 years 2 months)

- Competitive ML practitioner with top-tier performance in large-scale Kaggle competitions.
- Specialized in model development, evaluation, and optimization for computer vision and structured prediction tasks.
- Mentored practitioners and contributed to open ML communities through kernels, discussions, and code.
- Developed expertise in experimentation, feature engineering, model evaluation, optimization, and leaderboard-driven iteration under competitive constraints.

## GitHub

### Independent Software Architect (Data & ML)

June 2013 - April 2016 (2 years 11 months)

Greater Bengaluru Area

- Worked as a technical program manager and software architect on data-intensive and ML-driven systems.
- Contributed to and led open-source projects, collaborating with distributed engineering teams.
- Built and evaluated machine learning models for real-world computer vision and NLP applications.
- Built foundations in distributed software engineering, open-source collaboration, data systems, and machine learning applications.

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## Education

### Anna University Chennai

Bachelor of Technology - BTech, Information Technology · (2009 - 2013)

### Indian Institute of Management Bangalore

Online Certificate Courseworks - IIMBx, Entrepreneurship/Entrepreneurial Studies · (2020 - 2020)

### Stanford University

Independent Coursework, Artificial Intelligence, Machine Learning, Deep Learning for NLP and CV, Robotics with Deep RL

### Massachusetts Institute of Technology

Independent Coursework, Algorithms and Programming, Deep Learning for Self Driving Cars, and Artificial General Intelligence

### Harvard University

Independent Coursework, Computer/Information Technology Administration and Management